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# Network Analysis

## Third Assignment

# 1 Overview

The purpose of this simulation is to evaluate **how networks behave under different types of attacks**, aiming to assess their **robustness** and explore potential strategies to **improve resilience**.
We consider three types of networks:

- **Erdős–Rényi (ER)**: A synthetic random network model.
- **Barabási–Albert (BA)**: A synthetic scale-free network model.
- **Facebook Politician Network**: A **real-world social network**.

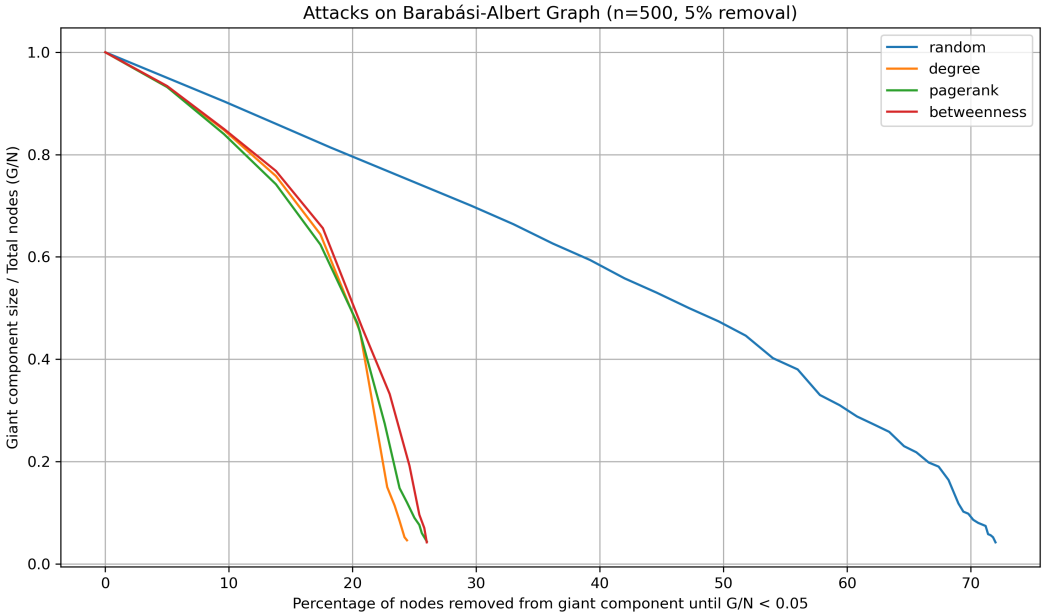
Each network is subjected to the following **attack strategies**:

- **Random**: Nodes are removed at random.
- **Degree-based**: Nodes with the highest degree are removed.
- **PageRank-based**: Nodes with the highest PageRank scores are removed.
- **Betweenness-based**: Nodes with the highest betweenness centrality are removed.

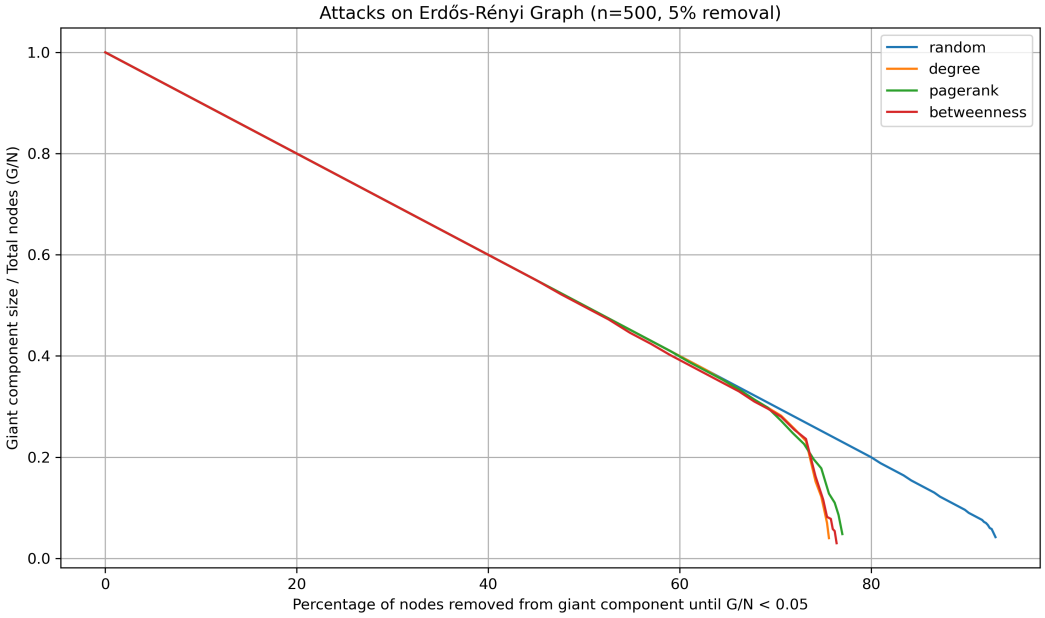
All attacks are applied by removing **5% of the nodes** in the **giant connected component** during each iteration. The simulation continues until the **G/N** becomes lower than **0.05**.
The primary goal is to understand **how long the network can maintain connectivity**, that is, to observe how many steps it takes for the giant component to shrink below a critical threshold. This allows us to compare the **resilience of different network topologies** under various attack types.

# 2 Synthetic Network

## Barabási–Albert



# Erdős–Rényi



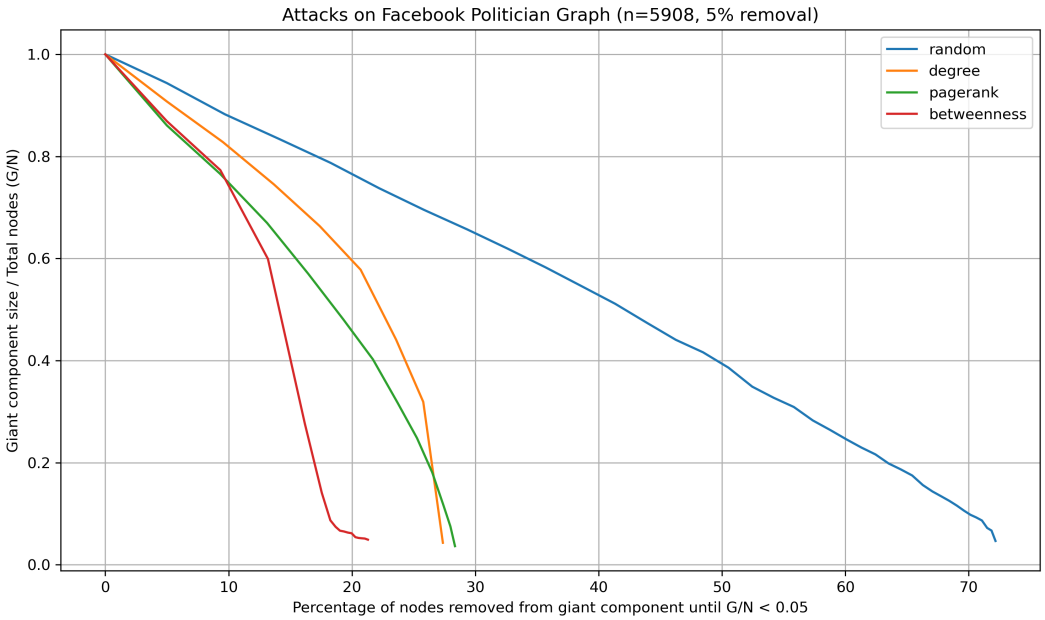

# Summary Table

| Graph | Strategy | Steps to G/N < 0.05 | Percentage Removed (%) |
|---|---|---|---|
| Erdős–Rényi | Random | 61 | 93.0 |
| | Degree | 31 | 75.6 |
| | PageRank | 33 | 77.0 |
| | Betweenness | 34 | 76.4 |
| Barabási–Albert | Random | 38 | 72.0 |
| | Degree | 10 | 24.4 |
| | PageRank | 13 | 26.0 |
| | Betweenness | 10 | 26.0 |

The **Erdős–Rényi (ER) network** is generally **more robust** than the **Barabási–Albert (BA) network** against all types of attacks. This is because the ER network has a more **random structure** with **fewer important hubs and key nodes**, making it less vulnerable.
For example, under **random attacks**, the ER network lasts for **61 steps** and requires removing about **93% of nodes** to break down, while the BA network fails after **38 steps** with only **72% of nodes removed**. In **degree-based attacks**, the ER network holds up for **31 steps** removing around **76% of nodes**, but the BA network collapses after just **10 steps** with only **24% of nodes removed**.
**Betweenness-based attacks** show an even bigger difference: the ER network resists **34 steps** with **76% of nodes removed**, while the BA network breaks after **10 steps** with just **26% removed**. This clearly shows that the ER network is stronger because its structure spreads connections more evenly, making it harder to disrupt by removing key nodes.

# 3 Real Network

## Facebook Politician Network




In our real-world network, **all targeted attacks are highly effective**. Among these, the **degree-based attack** is the most impactful, as shown in the table below. With just **8 iterations**, it reduces the network below the critical threshold of $G/N < 0.05$. It is also the **fastest in terms of execution time**, requiring roughly **0.67 seconds**, which is significantly less than the next most efficient method, **PageRank** (about **1.06 seconds**). This remarkable efficiency arises because the network depends heavily on **hubs** to maintain connectivity. Removing the top 5% highest-degree nodes at each step causes the **network structure to collapse rapidly**.

On the other hand, the **betweenness-based attack** requires roughly twice as many iterations (16 steps) but, interestingly, has a slightly **faster percentage removal** (around **21.3%**) compared to degree (**27.4%**), indicating a **more targeted disruption**. However, the **execution time is significantly longer** (**401.36 seconds**), which is expected since betweenness centrality requires recalculating shortest paths at every iteration. Consequently, as the network size grows, this attack becomes increasingly inefficient in practice, despite its theoretical effectiveness.

## Facebook Politician Network

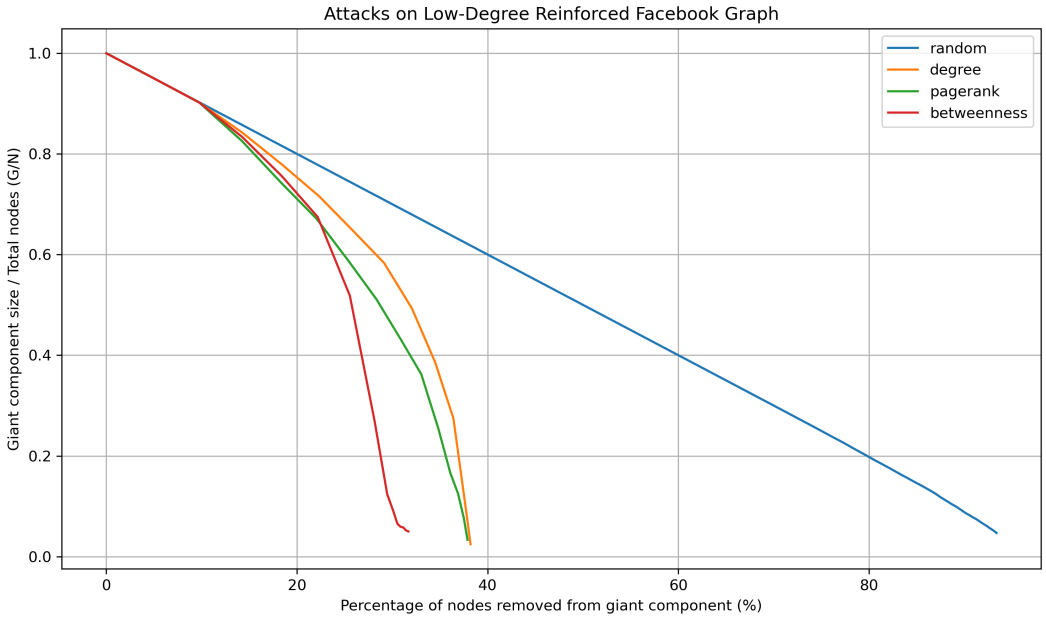
| Strategy | Steps to G/N < 0.05 | Percentage Removed (%) | Execution Time (s) |
|---|---|---|---|
| Random | 38 | 72.17 | 1.98 |
| **Degree** | **8** | **27.37** | **0.67** |
| PageRank | 12 | 28.35 | 1.06 |
| **Betweenness** | **16** | **21.29** | **401.36** |

# 4 Improving the Real Network

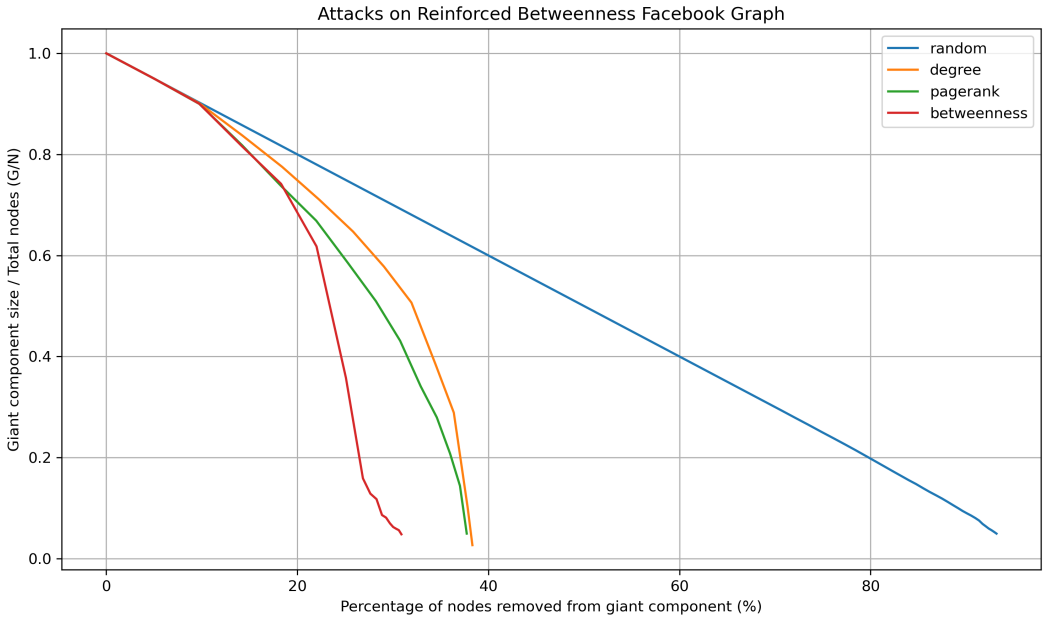
To improve the network's resilience against previous attacks, I used two approaches based on the insights gained.
**The first approach focuses on low-degree nodes:** We know that **removing hubs causes the most damage, as they are often the nodes that connect the majority of giant component**, so the idea was to **increase the number of edges by 3% for each of the 10% of nodes with the lowest degree** within the giant component. This creates more connections between nodes, **reducing dependency on hubs**, so even if a hub is removed, the **giant component remains intact**.
**The second approach is similar but based on low betweenness centrality:** In this case, the goal is to **increase the betweenness of the weakest nodes** by **adding 3% of new edges for each of the 10% of nodes with the lowest betweenness**. As explained earlier, this aims to **enhance the system's resilience** by creating additional alternative paths within the network.

**The choice of applying these modifications only to 10% of the nodes and limiting the increase to 3% of the total size of the giant component** is motivated by the fact that, as more nodes or edges are added, the **global properties of the graph tend to shift progressively**. By keeping these values low, we aim to **minimize structural alterations**, preserving the overall characteristics of the original network while still improving its robustness.

## Facebook Politician Network — Reinforcement of Low-Degree Nodes



# Facebook Politician Network — Reinforcement of Low-Betweenness Nodes




## Facebook Politician Network - Attack Comparison

| Graph | Strategy | Steps to G/N < 0.05 | % Node Removed | FC Before | FC After | Edges |
|---|---|---|---|---|---|---|
| Original | Random | 38 | 72.17 | 0.976 | | 41729 |
| | Degree | 8 | 27.37 | | | |
| | PageRank | 12 | 28.35 | | | |
| | Betweenness | 16 | 21.29 | | | |
| Reinforced Low Degree | Random | 56 | 93.36 | | 0.990 | 146159 |
| | Degree | 12 | 38.20 | | | |
| | PageRank | 14 | 37.88 | | | |
| | Betweenness | 14 | 31.69 | | | |
| Reinforced Low Betweenness | Random | 55 | 93.18 | | 0.990 | 146159 |
| | Degree | 12 | 38.32 | | | |
| | PageRank | 13 | 37.73 | | | |
| | Betweenness | 16 | 30.89 | | | |

As we can see from the graph, **both reinforcement modifications have brought improvements**. The most noticeable change is in the **random attack**, which with the two algorithms increased from **39 to 55/56 iterations**, accompanied by a substantial rise in the **percentage of nodes removed** from about **72% up to over 93%**. There is also a significant increase in the **degree attack**, with an increment of **4 iterations** in both cases and a corresponding increase in nodes removed from roughly **27% to around 38%**. The only value that showed a **decrease in iterations** is the **betweenness attack** in the **low degree graph**, likely because adding new edges **increased betweenness in the graph**, thereby indirectly enhancing the attack's impact. The percentage of nodes removed in this case decreased slightly from about **31.7% to 30.9%**.
Both solutions managed to **raise the critical threshold from 0.976 to 0.990**, which is an **excellent result**. **However, all these improvements came at the cost of tripling the number of edges in the network**, going from **41,729** in the original graph to **146,159** in the reinforced versions.

# 5 Conclusions

Considering all the previously analyzed networks, I can conclude that **giant components are generally vulnerable to targeted attacks** based on measures such as **degree, PageRank, and betweenness centrality**, as these nodes are responsible for maintaining the highest number of connections within the network.
In the specific case of my graph, it already started with a **high critical threshold** (around 0.976), which is a **positive indicator of resilience**: removing individual nodes does not significantly affect the giant component. Problems arise when **multiple nodes are removed simultaneously**, introducing **greater instability**.
The **reinforcement strategies implemented** have significantly increased the number of iterations required to reduce the giant component below the critical state (e.g., from 38 to 55–56 iterations for random attacks). Similarly, **notable improvements can be observed for other important strategies**, such as degree and PageRank attacks, in terms of both iterations and percentage of nodes removed to reach the critical threshold.
They also **raised the critical threshold further** (up to 0.990), which is notable given the network's already high robustness.
In conclusion, we can say that the network was **inherently robust from the beginning**, so such reinforcement strategies might be **unnecessary under normal conditions**. It is unlikely that so many nodes would fail simultaneously. Moreover, **adding edges should be carefully considered**, as it can alter the network's internal properties.
However, if the goal is to **maximize robustness at all costs**, even at the expense of some original structural features, then **both reinforcement strategies have proven effective**. Of the two, the **low betweenness strategy requires more computational time** due to repeated shortest path calculations, but it also **achieves the best overall performance**.